AVINASH MISHRA
Manager (Data Scientist) | Generative AI & LLM Specialist | AI Agent Systems Architect
Email: avinash.mishra.2388@gmail.com | Mobile: +91-7909409790
LinkedIn | GitHub | StackOverflow | Portfolio | HackerRank

Experienced AI leader with hands-on expertise in building data science workflows, AI-agent systems, RAG pipelines, VLM-powered document understanding, and responsible AI governance. Proven success in deploying enterprise-scale AI products (chatbots, AI assistants, regulatory AI, policy GPTs) serving millions of users. Currently leading a team of 17 members across data science, statistics, engineering, BA, and product functions.

ACHIEVEMENTS & CERTIFICATIONS

- Oracle Certified GenAI Professional
- StackOverflow Medals: 4 Gold, 30 Silver, 44 Bronze
- Data Science Talks:
  - YouTube: "Deploying Scalable ML models using K8s and TFserving/FlaskAPI"
  - Odin Talks: "Data Science Banker: Insights and Innovation in Finance"
  - Tech Talks: 3 presentations at Global IT Centre SBI, Navi Mumbai. *Recent was about cognitive memory for AI-Agents.*
- Founder: Data-Science-Trend
- Author: "Data Science Essentials – An open-source iPy-Book"
- Speaker: International Star-Conference: "Neural n/w approaches for recommendation engine"
- Excellence Certificate: Certificate of Excellence for askSBI chatbot, the first enterprise-scale AI Agent at SBI.
- **ETHR Award 2025** for Best Advance in Generative AI Solutions (askSBI chatbot).
- **BFSI Exceller Award 2025** for building Generative AI model in hybrid architecture**.**
- Technical Committee member to setup SBI Foundation Hub for Data Science and Analytics with IIT-Bombay**.**
- Winner: Hacktoberfest 2019 (out of 154,466 participants)
- Analytics Journal award: Recognized multiple times with the *SBI Analytics Journal Award* for the ability to distil complex data insights into clear, actionable presentations for non-technical stakeholders.

EXPERIENCE

State Bank of India (SBI), India | Manager (Data Science) | Mar 2021 – Present

- Designed and implemented **VMA (Vision, Mission & Action) plan** to align Analytics strategy with business goals.
- Led development of **scalable Data Science solutions** impacting 500M+ customers.

- Led the initiative of creating and enforcing **Generative AI governance policy** for responsible AI use.
- Led transformation of Analytics department into an **AI Centre of Excellence**, defining KRAs and role structures.
- Directed **cross-functional collaboration**, project prioritization, and CUG testing for business impact.

Business Growth Models

- *Customer-DNA and lifetime value modeling (CDNA-CLV):* Umbrella project for hyper-personalization, omnichannel strategy and long-term customer engagement providing 360° personas of 24K+ branches and 500 Million customers. Patent file initiated for internal WCPPM algorithm. Total advances: 81.9 Billion INR, 2.8% conversion rate
- *Pre-approved Traders Loan*: Product targeting Savings Account holders doing small value UPI transactions to offer low value Business Loans under SME segment. Techniques: Cross-Tab, RFM, clustering and RSM. Total disbursement FY23: 5.42 Billion INR, 21.3% conversion rate
- *Next Best Product*: Neural network-based product recommendation for CSP customers

GenAI Models

- askSBI GenAI Chatbot: Award-winning AI Agent system serving 250K internal users, designed with RAG-based retrieval, hybrid cloud deployment (Azure + Private Cloud), and enterprise-grade scalability.
- Regulatory GPT, SOP-GPT, Deceased Account Settlement Bot, Knowledge Bot for Project finance, Contact Centre Assistant: Developed and deployed a suite of AI Agent solutions with agent orchestration, safe AI deployment practices, and domain-specific compliance checks.
- AI Governance & Safety: Designed and implemented AI governance policies for hallucination control, fact-checking, citation enforcement, and responsible deployment of LLMs in financial services.
- Benchmarking & Continuous Evaluation: Built and maintained task-specific benchmarks for financial GenAI agents, enabling continuous monitoring, evaluation, and improvement of model accuracy, safety, and business impact.

Responsible AI Framework

- Designed and launched **Responsible AI framework (R-AI)** for safe deployment of LLMs in banking.

Risk, Fraud & NPA Management

- Fraud prevention model for branches: Time series and Change point detection model to identify outlier branches to prevent fraud, reducing false positive alerts
- Internal Risk Scoring of customers: Blended model using rule-based plus RandomForest to provide risk scoring for all SBI customers (0-100 range)
- NMI Predictor: Used XGBoost to predict NMI of CSP and non-CSP customers
- NTC model: For new-to-credit customers, an internal score (0-4 range) is predicted using clustering

- Best Time to call a customer: Multi-label classification model predicting top 2 slots to call stressed SBI customers using LAMS data from contact center

Rakuten, Japan | Data Scientist | Dec 2020 – Feb 2021

- Senior Member of Insight Analytics Group in Data Science and Consulting Department
- Developed 'at home prediction model' for Rakuten Ichiba to optimize last-mile delivery attempts using CatBoost classifier
- Built multi-point data fetcher package enabling seamless integration of various Rakuten data sources within Jupyter notebook

Catalina Marketing Japan (CMJ), Japan | Data Scientist | June 2018 – Nov 2020

- Led a **multinational Data Science team**, delivering AI/ML solutions for **retail and CPG clients** across in-store and omnichannel platforms.
- Expanded team capabilities, initiated **monthly Data Science "Shimekai" program**, and managed **internship & cross-country collaborations** (US, France, Germany).

Key Projects

- New Product Launcher: Solution for MFG clients to track and forecast sales of newly launched products. Led project and built ML-based forecaster and web app Dashboard using Plotly's Dash framework. Successfully piloted for Suntory and Somi MFG in Japan.
- Similarity at scale (ID mapping): Implemented Locality Sensitive Hashing using Spark to identify matching entities across datasets. Collaborated with Cyberagent, BlogWatcher, and Mobilewala.
- Media Mix Tracker (MMT): Statistical analysis tool using multivariate regressions to estimate marketing tactics' impact on sales. Used OLS model for descriptive analysis and LightGBM/NN blend for prediction with SHAP explanations. Piloted for Mizkan and Suntory.
- Likely buyer: Classification model to identify potential buyers for marketing campaigns, achieving ~17% higher conversions.
- Similar Basket: Product similarity model using word2vec to find similar products across baskets in retail chains.
- Learning to reRank (LINE coupons): Coupon ranking system using binomial distribution over CTR to optimize offer placement in LINE app.
- Customer Segmentation: Automated pipeline to segment 90 Million monthly transactions based on shopping behaviors.
- Store Revenue booster: Sales prediction model for retail managers with 7-day, 15-day, 30-day, and 90-day forecasts.
- Retail_AutoML: Built autoML platform specific to retail and CPG industry datasets supporting regression and classification.

Paradigmshift.io, Japan | Software Engineer (Data Science Lead) | Sept 2016 – May 2018

- Led multinational Data Science team building state-of-the-art NLP systems
- Reported to COO on existing projects progress and new initiatives

- Trained team to build large-scale microservice-based Data Science systems
- Acquired 25 million JPY budget to expand AI team, building it from scratch

Key Projects

- RepChecker: NLP tool providing insights into OTA reviews for hotels and ryokans in Japan. Designed ML architecture implementing various NLP techniques supporting 5 languages.
- Price RecSys for RCBooking: Price recommendation engine for OTA price master to hotel managers optimizing for 100% room occupancy.
- Smart web crawler: Designed distributed web crawling architecture with centralized controller, distributed spiders, and Cassandra cluster storage.

Rakuten, Japan | Software Engineer | Oct 2014 – Sept 2016

- Leads for Rakuten Pink card for women: Built Xgboost based ML model to identify potential leads for Rakuten Pink card for Japanese women.
- Card transaction categorization: Utilize NLP techniques to process card transaction and assign it to categories like travel, groceries, utilities etc.

Infosys Ltd., India | Systems Engineer | Mar 2011 – Aug 2012

- Molina Healthcare management: Managed users' health information database, designed SQL Server database with stored procedures, triggers, and functions.
- Employee SWAP portal: Designed web-based portal for employees to swap posting locations, streamlining transfer processes.

EDUCATION

- **Master of Technology in Software Engineering**, VJTI Mumbai (2012–2014)
    - Teaching Assistant: Data Structures & Algorithms, Software Engineering
- **Bachelor of Engineering in Computer Science**, RGPV Bhopal (2006–2010)

TECHNICAL SKILLS

- **Languages**: Python, SQL, Bash, Scala, R, Java, JavaScript, PHP
- **LLM/GenAI Frameworks:** LangChain, LlamaIndex, Haystack, SmolAgents, AutoGen, CrewAI, OpenAI GPT, Gemini, vLLM, Ollama, HuggingFace Transformers.
- **AI Agent Tools:** MCP, A2A, function calling, prompt orchestration.
- **RAG & Search:** CosmosDB, ChromaDB, Weaviate, FAISS, ElasticSearch, Solr, Apache Lucene
- **Multimodal AI:** VLMs, OCR (Tesseract, LayoutLM, Donut).
- **Responsible AI:** XAI, fact-checking pipelines, hallucination guardrails, citation injection.
- **Data Science**: NLP, NLTK, Scikit-Learn, Keras, TensorFlow, PyTorch, Spark MLlib, MLflow, PyCaret, Prophet, GenAI, RPA
- **Data Visualization**: Tableau, DOMO, Matplotlib, Kibana, Dash by Plotly,
- **Web Frameworks**: Django, Flask, Laravel, VueJs, Node.js, FastAPI

- **Big Data Technologies**: HDFS (HDP), Hive, MapReduce, Spark, Kafka, Zookeeper, Ambari, AirFlow
- **Servers**: Apache Tomcat, Jetty, Nginx
- **Databases**: MySQL, Cassandra (Apache, DSE), MongoDB, MariaDB, SQL Server, Redis, Netezza, IBM DB2, CP4D, Oracle DB
- **DevSecOps/CI/CD**: Jenkins, Chef, Docker, Vagrant, Apache Maven, Kubernetes
- **Tools**: GitHub, Jira, Trello, Confluence, Bitbucket, GitLab, Target Process, SAS
- **Cloud Platforms**: AWS, DigitalOcean, Amazon Lightsail, Azure Cloud, GCP

SBI AWARDS/GEMs (LAST 4 YEARS)

- 15+ recognitions, including **2 from Deputy Managing Directors, 2 from Chief General Managers, 1 from GM, 3 from DGMs, and 8 from AGMs** for excellence in AI/Data Science initiatives.

LANGUAGES

- English (Business)
- Hindi (Native)
- Japanese (N4, Basic Conversational)

INTERESTS

- Playing Badminton (State-level player in India)
- Reading technological news, trends and exploring AI-related GitHub repositories